

Personalized bacteriophage therapy for chronic biliary tract *Pseudomonas aeruginosa* infections

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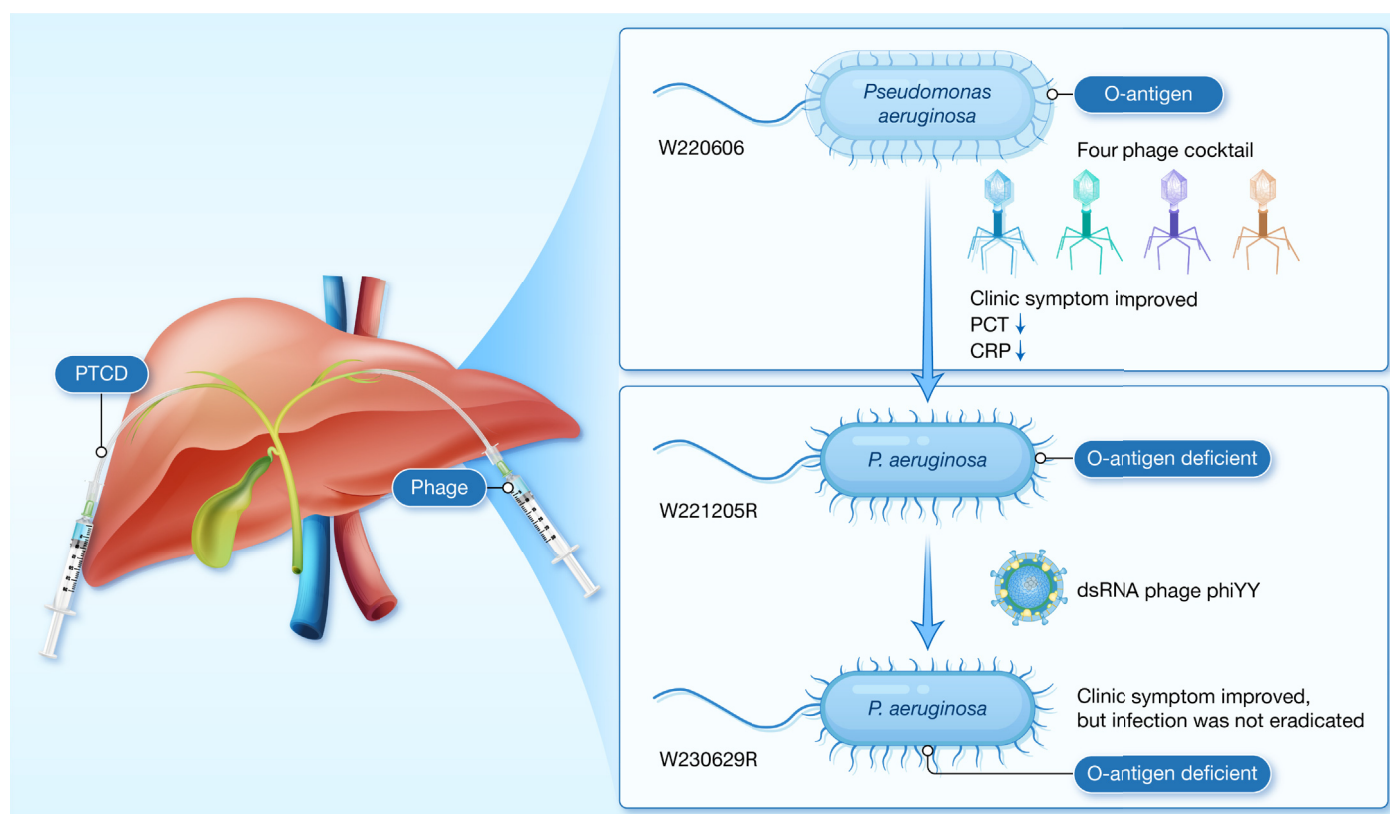
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GRAPHICAL ABSTRACT



HIGHLIGHTS

- Phage delivery through percutaneous transhepatic cholangiodrainage (PTCD) effectively controls biliary infection.
- Phage therapy selects for O-antigen-deficient resistant mutants.
- The dsRNA phage phiYY suppresses but fails to clear O-antigen-deficient bacteria.



Personalized bacteriophage therapy for chronic biliary tract *Pseudomonas aeruginosa* infections

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ABSTRACT

Biliary tract infections (BTIs) present a significant therapeutic challenge, particularly in the face of increasing antimicrobial resistance. Bacteriophages, viruses that target and destroy bacteria, offer the potential for treating severe bacterial infections, although their use in BTIs has been limited. We describe an 88-year-old female with a complex and recurrent BTI caused by multiple bacteria, including multidrug-resistant *Pseudomonas aeruginosa*. Despite treatment with various antibiotics and percutaneous transhepatic cholangiodrainage (PTCD), her condition did not improve. As a final measure, we implemented personalized phage therapy in combination with antibiotics. An initial 9-day antibiotic treatment combined with a *P. aeruginosa* phage cocktail administered via PTCD fluid resulted in significant symptom relief. However, phage-resistant pathogens emerged, exhibiting resistance to all 100 double-stranded DNA (dsDNA) phages in our library due to genetic mutations affecting lipopolysaccharides biosynthesis. A second round of therapy with a double-stranded RNA (dsRNA) phage, phiYY, which targets O-antigen deficient mutants, was subsequently administered. Although complete eradication of *P. aeruginosa* was not achieved, the patient's clinical symptoms were markedly improved. This case demonstrated the safety and efficacy of phage therapy in the treatment of BTIs and showcased the feasibility of employing dsRNA phages to combat the emergence of O-antigen-deficient bacterial mutants. However, it also underscores the considerable challenges in completely eradicating persistent *P. aeruginosa* infections, which may be attributed to bacterial heterogeneity, biofilm formation, and phage-resistant genetic mutations.

KEYWORDS phage therapy; biliary tract infections; *Pseudomonas aeruginosa*

INTRODUCTION

Biliary tract infections (BTIs) frequently lead to bacteremia and are correlated with high rates of morbidity and mortality. Common symptoms include pain, fever, and jaundice, along with elevated inflammatory markers and abnormal liver func-

tion tests [1,2]. There is a strong link between BTI and structural abnormalities, such as choledocholithiasis and malignancies, which may account for up to 69% of cases [3]. Moreover, the emergence of drug-resistant bacteria has significantly reduced the effectiveness of antibiotic treatment

[4,5]. While Enterobacteriaceae are the most common causes of BTIs, *Pseudomonas aeruginosa*, though less prevalent, poses a more challenging treatment due to its antibiotic resistance and propensity for biofilm formation, resulting in chronic infections [6].

Lytic bacteriophages are recognized for their significant therapeutic potential in combating severe bacterial infections [7,8]. Over recent years, the clinical deployment of phage therapy has expanded considerably, demonstrating exceptional promise in countering infections from multidrug-resistant bacteria [9–11]. The paradigm of personalized medicine, with treatments specifically customized to the pathogen afflicting an individual patient, is steadily advancing from concept to clinical reality [12,13]. Phages have been applied to treat various infections, such as respiratory tract infections, bone and orthopedic prostheses infections, skin and soft tissue infections, bloodstream infections, urinary tract infections, and other infection types [8,14]. The clinical merits of personalized phage therapy are underscored by its proven effectiveness against infections caused by antibiotic-resistant bacteria and by its versatility in being adapted across a spectrum of clinical settings [7]. However, several limitations of phage therapy impede its more extensive clinical application. These include the development of phage resistance, the host-specific nature of phages, the immune response they trigger, the scarcity of commercialized phage products, and the absence of a standardized protocol for phage therapy [14–16]. Thus, more basic and clinical investigations are required to facilitate its applications.

Moreover, phage therapy in treating BTIs has not been reported yet, to the best of our knowledge. In this study, we present the case of an 88-year-old female with a four-year history of complex and recurrent BTIs. She had undergone multiple percutaneous transhepatic cholangiodrainage (PTCD) interventions and had received prolonged treatment with conventional antibiotics, all of which failed to achieve a cure. We detail the clinical and therapeutic aspects of her case, underscoring both the challenges and opportunities that phage therapy presents for the treatment of BTIs. To the best of our knowledge, this is the first report on the use of phage therapy in treating BTIs. Our findings indicate that phage therapy led to a notable improvement in clinical symptoms, though it did not entirely eradicate *P. aeruginosa*, which may be attributed to factors such as bacterial heterogeneity, biofilm formation, and/or the emergence of phage-resistant mutants.

RESULTS

An 88-year-old female with a recurrent history of cholesterol gallstones has faced a series of complex and recurrent BTIs caused by various bacterial pathogens. Since 2019, she has been battling persistent symptoms, such as recurrent fevers, abdominal pain, and jaundice. Despite numerous treatments with a range of antibiotic therapies and continuous PTCD, a cure has remained out of reach. Additionally, cholelithiasis and ongoing inflammation have led to worsening structural abnormalities in her bile ducts, as confirmed by magnetic resonance imaging (MRI) and ultrasonography.

Following prolonged antibiotic therapy, the BTI has evolved into a chronic inflammatory condition. During this time, MRI

scans hinted at the possible presence of a bile duct carcinoma, which was later confirmed as a low-grade malignant tumor via biopsy. Nonetheless, the patient and her family chose a conservative treatment strategy.

In June 2022, the patient presented with acute high fever and abdominal pain. Bile culture identified a multidrug-resistant isolate of *P. aeruginosa*. An initial course of treatment with meropenem and levofloxacin for 14 days was successful. However, the infection recurred within two months, showing resistance to ceftazidime and subsequent oral cefixime therapy. By November 2022, she experienced increasing abdominal pain, and *P. aeruginosa* was once again detected in her bile, with resistance to carbapenems and quinolones, and intermediate resistance to ticarcillin/clavulanic acid, cefoperazone/sulbactam, and aztreonam. She was admitted to our department for the management of her complicated BTIs. Considering the patient's documented allergies to penicillin, sulfonamides, and amikacin, along with inadequate peripheral venous access and her reluctance to indwell a central venous catheterization, the therapeutic options were constrained. Faced with a scarcity of effective antibiotics, the patient voluntarily gave her informed consent to commence phage therapy (Figure 1).

The First Course of Phage Therapy with a 4-phage Cocktail

Drawing upon the successful identification of bacteriophages with efficacy against the mutiantibiotic-resistant *P. aeruginosa* strains isolated from the patient's bile (Figure 2A), we proceeded to devise a customized bacteriophage cocktail. This cocktail, comprising phages ϕ PA-A60, ϕ PA-A69, ϕ PA-A70, and ϕ PA-A78, which all efficiently lyses strain W220606, was specifically tailored for the initial round of phage therapy (Figure 2B). The formulation and preparation of these good manufacturing practice (GMP)-compliant phage products were executed as described [14].

Given the lack of established pharmacological data regarding bacteriophage administration in BTIs, and in the absence of a unified medical consensus on intrahepatic delivery methods, a multidisciplinary consultation was convened. The gathering of medical professionals from relevant fields aimed to deliberate and refine the treatment strategy. Employing cholangiography for a precise assessment, we devised an initial administration plan for the phage therapy (Figure 3A). This plan allocated a volume of 5 mL for infusion into the left bile duct and 3 mL for the right, ensuring a targeted approach to treatment.

Before initiating the phage therapy, we performed bile extraction via the patient's PTCD tubes. Following this procedure, an equimolar concentration of the bacteriophage cocktail (10^8 plaque-forming units per milliliter [PFU/mL]) was carefully introduced into the bile ducts. The drainage tubes were clamped for 30 min to ensure optimal contact between the phages and the target bacteria. The phage therapy was administered for nine days, from November 28 to December 6, 2022 (Figure 1). Throughout this treatment phase, the patient was placed under strict and continuous monitoring to assess the therapy's safety and efficacy. Notably, after the second dose of phage application, the patient reported a brief episode of chills, which promptly subsided within approximately 30 min.

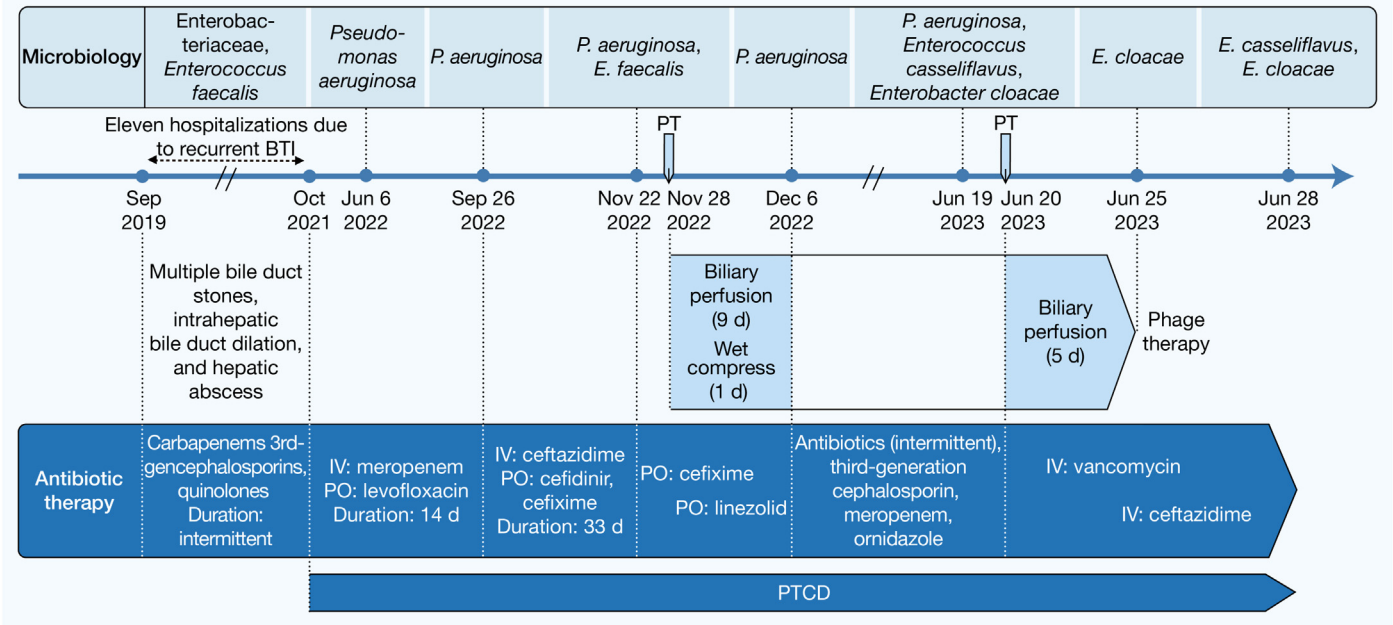


Figure 1. Timeline of the most relevant microbiological results, antibiotic therapies, and phage therapy
Two courses of phage therapy were performed with a 4-phage cocktail (φPA-A60, φPA-A69, φPA-A70, and φPA-A78) and a dsRNA phage phiYY, respectively. Abbreviations: BTI, biliary tract infections; PT, phage therapy; IV, intravenous; PO, per os; PTCD, percutaneous trans-hepatic cholangiodrainage; dsRNA, double-stranded RNA.

Bile cultures exhibited negative results on the first day (day 1) postinitial treatment (day 0). However, *P. aeruginosa* isolates were subsequently detected on days 3, 5, and 7. Along with the simultaneous detection of *Enterococcus faecalis* on day 3, the antibiotic used in combination had been replaced with linezolid instead of cefixime. While complete eradication of *P. aeruginosa* isolates was not achieved, the patient’s abdominal symptoms improved, and the bile color tended toward normalized (Figure 3B). Notably, inflammatory markers

of C-reactive protein (CRP) and procalcitonin (PCT) both exhibited a consistent decline over time (Figure 3C). Additionally, a total of two topical applications of the bacteriophage cocktail were administered to the skin surface at the right abdominal puncture site, effectively clearing the infection at the puncture site and thereby demonstrating the efficacy of bacteriophage therapy (Figure 3D). However, phage susceptibility tests revealed that *P. aeruginosa* strains isolated on day 5 and day 7 exhibited complete resistance to all double-stranded DNA

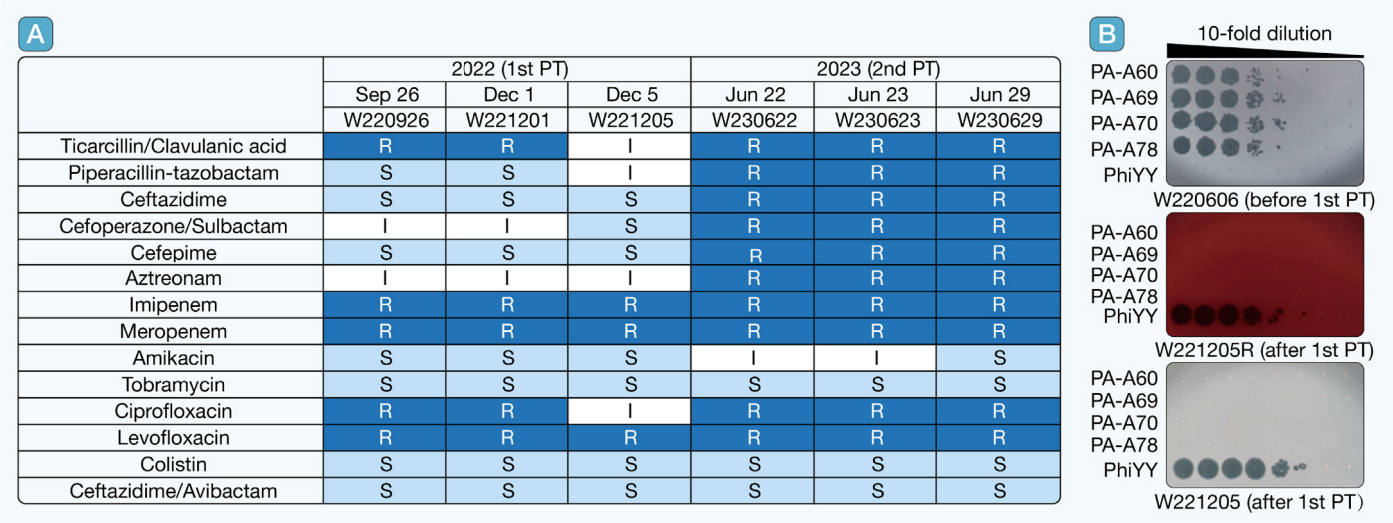


Figure 2. Antibiotic susceptibility and phage sensitivity profiles of representative bacterial strains
(A) Antibiotic susceptibility assay of the strains isolated from the bile before and after two courses of phage therapy. Strain W220926 was isolated before phage therapy, while the remaining strains were obtained during and after the treatment. (B) The efficiency of plating (EOP) assay to test the phage sensitivity of three strains isolated before and after the first course of a 4-phage cocktail therapy. Strains W221205 and W221205R developed resistance against the 4-phage cocktail, but became sensitive to dsRNA phage phiYY, and specifically, W221205R is a brown pigment-producing *P. aeruginosa* strain. Abbreviations: PT, phage therapy; EOP, efficiency of plating; R, resistant; I, intermediate; S, susceptible; dsRNA, double-stranded RNA.

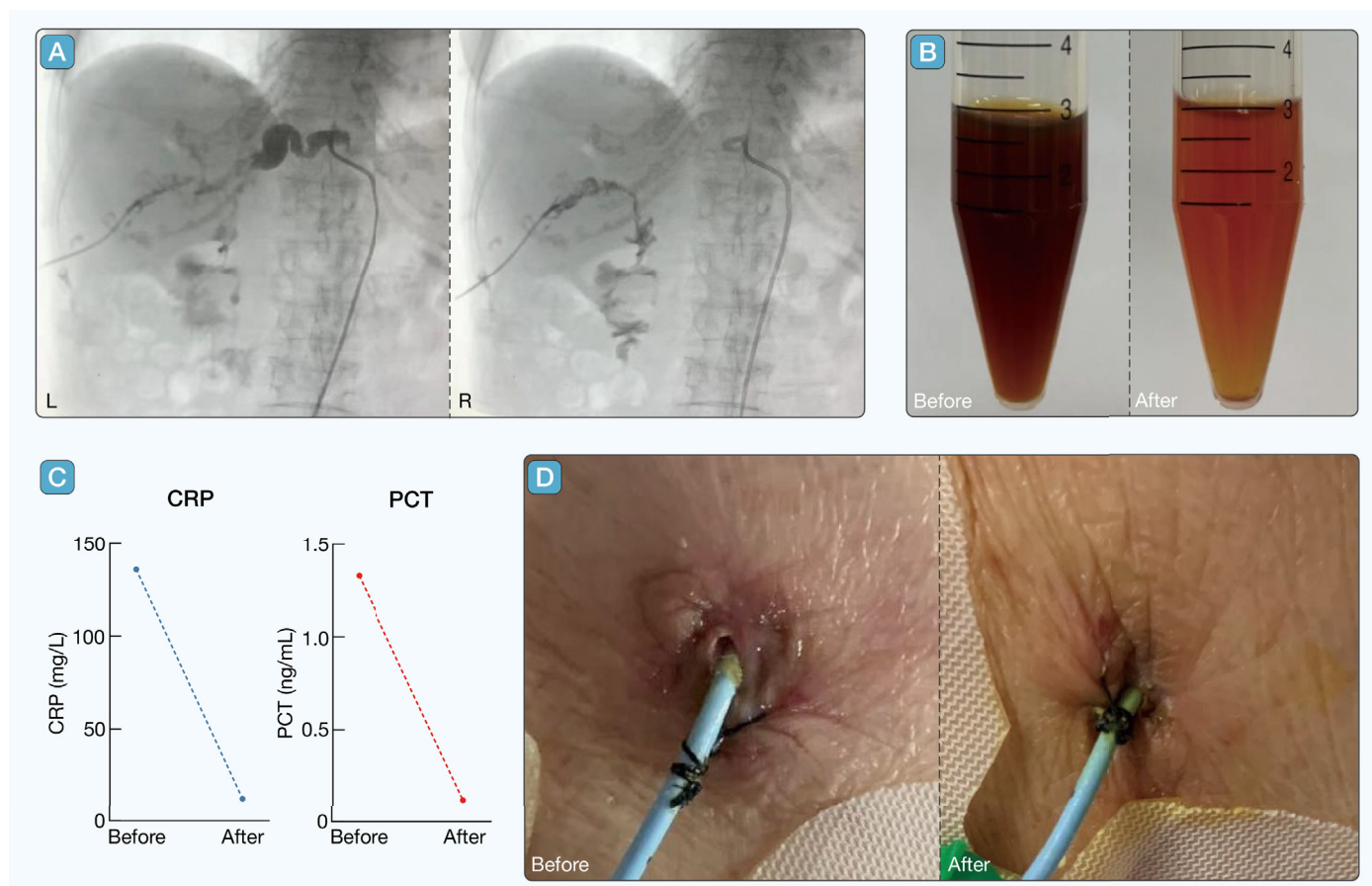


Figure 3. Evaluation of clinical outcomes associated with bacteriophage treatment

(A) Employing cholangiography to assess the volume for phage therapy. (B) The bile color tended toward normalization after the 1st phage therapy. (C) The inflammatory markers of CRP and PCT both declined after the 1st course of phage therapy. (D) A total of two topical applications of the bacteriophage cocktail effectively eliminated the wound infection at the right abdominal puncture site. Abbreviations: L, left bile duct; R, right bile duct; CRP, C-reactive protein; PCT, procalcitonin.

(dsDNA) phages available in our library. Intriguingly, these *P. aeruginosa* strains exhibited sensitivity to double-stranded RNA (dsRNA) phage phiYY (Figure 2B).

The Second Course of Phage Therapy with dsRNA Phage phiYY

Due to personal reasons of the patient, the second course of phage therapy was put on hold until June 2023. Before this, third-generation cephalosporin, meropenem, and ornidazole were intermittently used upon the detection of *P. aeruginosa*, *Enterococcus casseliflavus*, and *Enterobacter cloacae*. At the time of admission, the patient was experiencing viral respiratory infection, she had a fever of 38.5 °C and was administered orally with molnupiravir for 5 days.

Before proceeding with phage therapy, the host range of the newly isolated *P. aeruginosa* strain from the patient was thoroughly tested. It was found that the strain remained uniquely susceptible to the dsRNA phage phiYY. phiYY is a dsRNA phage with three segmented dsRNA genomes [17], which infects the O-antigen deficient *P. aeruginosa*, because it uses core-antigen as the receptor [18] and has been applied to treat the lung infection caused by O-antigen deficient *P. aeruginosa* [13].

Subsequently, phiYY was diluted in 10 mL of 0.9% NaCl (normal saline) to a titer of 10^8 PFU/mL and administered daily.

Additionally, the capacities of the left and right bile ducts were examined through cholangiography, revealing volumes of 5 mL and 2 mL, respectively.

The duration of phage treatment lasted for 6 days (June 20 to 25, 2023), with combined intravenous application of ceftazidime and vancomycin due to mixed infections. On June 23, the patient's temperature returned to normal. After the initial phage treatment (D0), bile cultures were consecutively conducted for five days; however, *P. aeruginosa* was stubbornly detected in those cultures (days 1–3, day 5) along with co-occurrence of *E. cloacae*, *E. casseliflavus*, and *Candida albicans*. Nevertheless, the bacterial load had decreased slightly from 3+ to 2+ according to the semi-quantitative clinical cultures approach to measure the bacterial load [13]. Accordingly, the patient felt much better and was discharged on day 9.

Genome Analysis of *P. aeruginosa* Isolates Illustrating the Heterogeneity and Phage Resistance Mutations

We are particularly interested in linking the genomic mutations with the outcomes of phage therapy to guide our future treatment plans. Firstly, we want to know the phage-resistant mechanism generated during DNA phage therapy. Secondly, we are curious why phiYY failed to clear all the sensitive strains, and all the post-treatment isolates remained sensitive to the dsRNA phage phiYY.

Eight isolates, selected based on their distinct phage susceptibilities, were sequenced. Specifically, the W220606 isolate was subjected to both third-generation and next-generation sequencing techniques to produce a complete genome of 6,533,470 bp in size. The other isolates were analyzed through

next-generation sequencing and alignment with the reference genome of W220606, which uncovered a variety of single nucleotide polymorphisms (SNPs) and genomic deletions spanning from 1 bp to 285,287 bp (Figure 4A, Table S1). It is important to note that none of these strains harbored mutations in their

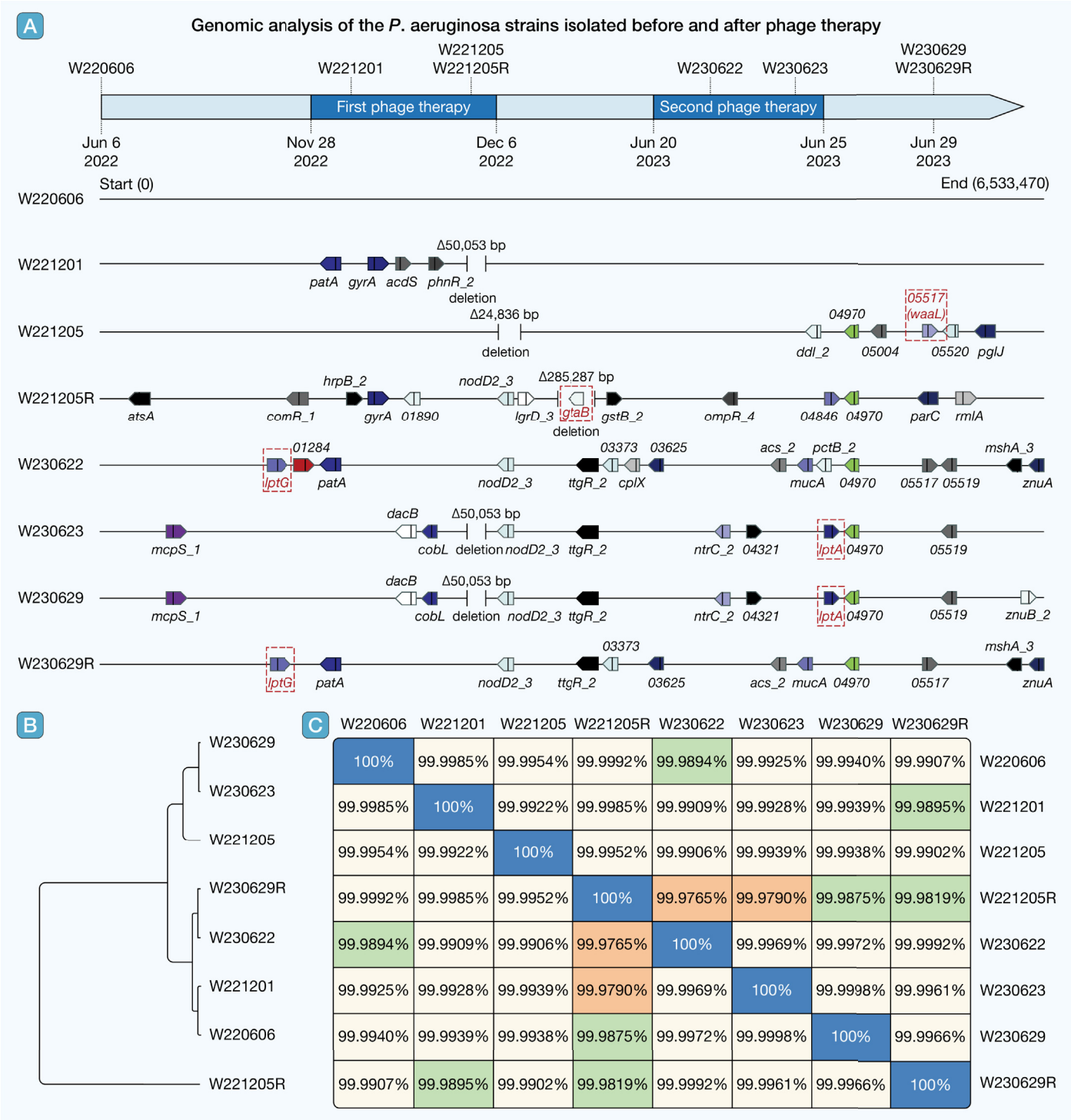


Figure 4. Genomic analysis of the isolated strains before and after phage therapy
(A) W220606 was used as a reference genome to identify the SNPs and genomic deletions present on seven other strains of *Pseudomonas aeruginosa*. The key genes highlighted in the red box are those involved in LPS biosynthesis which might affect phage sensitivity. (B) The genome-wide cladogram of eight sequenced *P. aeruginosa* isolates is based on the genome similarity calculated by Sourmash. (C) The heatmap of the pairwise similarities between the genomes of the eight sequenced *P. aeruginosa* based on ANI values indicates that these strains are generated from the same progenitor strain. Abbreviations: SNP, single-nucleotide polymorphism; LPS, lipopolysaccharide; ANI, average nucleotide identity.

mismatch repair systems, thus these strains have mutations within a range of 6–20 SNPs or deletions. The average nucleotide identity (ANI) values, calculated across the genome aggregate, exceeded 99.99%, indicating that these isolates are a group of clones derived from a single ancestral genome and can be classified as the same strain. The progenitor strain has been undergoing evolution for over one year, generating a multitude of heterogeneous clones that are selected during treatment (Figure 4B–4C).

W221201, isolated 4 days post-initiation of the phage therapy, retained sensitivity to the phage cocktail and was resistant to dsRNA phage phiYY. Its genome sequencing revealed 5 SNPs and a 50,053 bp deletion in a genomic fragment spanning from *orf02490* to *orf02552*. None of these mutations are implicated in lipopolysaccharide (LPS) biosynthesis (Table S1), indicating the heterogeneity of the bacterial populations at the site of chronic infection.

Strains W221205 and W221205R, isolated 8 days post-therapy initiation, exhibited complete resistance to all dsDNA phages in our library yet became sensitive to dsRNA phage phiYY. Notably, W221205R produced brown pigments on Luria-Bertani (LB) agar (Figure 2B), a result of a 285,287 bp genomic fragment deletion that includes the genes *hmgA* and *gtaB* (*galU*) (Table S1), leading to the accumulation of homogentisic acid and the loss of the O-antigen, respectively [10,19]. This phenomenon has been previously reported both *in vitro* and *in vivo*, which is generated spontaneously during bacterial replication [21], and these strains were selected by phage therapy. Isolate W221205 harbored 7 SNPs and a 24,836 bp genomic fragment deletion, including a mutation in *orf05517* (*waaL*), encoding an O-antigen ligase [22], resulting in the loss of the O-antigen and resistance to the dsDNA phage cocktail.

Following the first round of phage therapy, all strains developed resistance due to the loss of the O-antigen, prompting our decision to employ phiYY for the second round of therapy [18]. Sequencing of 4 isolates (W230622, W230623, W230629, and W230629R) from the phiYY treatment phase, conducted half a year later, revealed an increased number of mutations compared to the reference sequence, ranging from 12 to 20 SNPs, indicative of the ongoing evolution of these strains. Interestingly, all 4 strains presented mutations in either *lptA* or *lptG*, components of the LPS export system [23], yet remained sensitive to phiYY, suggesting they are O-antigen deficient strains.

This comprehensive genomic analysis has provided valuable insights into the dynamics of phage resistance and strain evolution in the context of *P. aeruginosa* infections, offering potential avenues for optimizing phage therapy strategies.

DISCUSSION

Effective management of BTIs generally depends on the administration of potent antibiotics, and the drainage of infected bile [24]. Nevertheless, the complexity of these infections often surpasses initial expectations due to various challenges, such as antibiotic resistance, allergic reactions to medications, and complications resulting from the use of implantable devices. Phage therapy, which was once overshadowed by traditional

antibiotics, is now being recognized as a promising adjunct for combating complex or antibiotic-resistant bacterial infections [25]. We present the case of a BTI patient with a complex *P. aeruginosa* infection of the biliary tract treated with a combination of phages and antibiotics. This treatment led to the resolution of abdominal signs and symptoms of infection, along with a reduction in bacterial load. We present the first protocol for phage therapy in BTIs, demonstrating its safety and efficacy, and providing valuable insights for future phage therapies in BTIs.

After the patient underwent the initial course of dsDNA phage cocktail therapy, the emergence of fully phage-resistant *P. aeruginosa* mutants was observed. These mutants were characterized by various mutations that result in O-antigen mutations, which may contribute to persistent infections and complicate management. For example, a 24,836 bp genomic fragment deletion was found in W221205R, including *galU* and *hmgA*. This deletion leads to the loss of O-antigen and the accumulation of homogentisic acid, respectively [20]. Additionally, a mutation in the LPS core region biosynthetic gene *dnpA* in W221205 might cause phage resistance during the first phage therapy. This indicates that phage therapy was very effective since the original dominant strain was eliminated, phage-resistant clones were selected, and the patient's symptoms were significantly alleviated.

Moreover, the O-antigen deficient strains appeared to be better adapted to survival in the chronically infected biliary tract, having dominated the population post-treatment and persisted for over six months. The O-antigen deficient strains had been frequently isolated from the sputum of the *P. aeruginosa* chronically infected cystic fibrosis patients and these strains might have fitness for chronic infections [21].

For the second course of dsRNA phage therapy, the patient was diagnosed with a viral respiratory infection during this period, complicating the interpretation of temperature and inflammatory marker changes. However, dsRNA phage therapy combined with antibiotics is effective in controlling the infection symptoms, though it failed to eliminate *P. aeruginosa*. All of the clones isolated during the second course of phage therapy exhibited various mutations in *lptA* or *lptG*, remaining sensitive to phiYY yet not eliminated. Further investigation is warranted to explore potential reasons, such as biofilm formation or bile toxicity to the lipid-enveloped dsRNA phages. Isolating dsDNA phages with better lytic activity and capable of infecting these LPS-minus mutants may present a viable alternative.

The challenges faced by phages in clinical practice include the emergence of phage-resistant strains, the heterogeneity and biofilm formation of clinical strains during chronic infection, and the potential limitations of the host immune response. Another possible explanation is the clonal replacement, wherein a new clone emerges or replaces the initial population. However, more isolates should be obtained and examined in the next phage therapy cases to prove this hypothesis.

These factors may plausibly account for the incomplete eradication of *P. aeruginosa*. Sequencing of 8 isolates revealed a high degree of similarity, yet each strain exhibited unique mutations.

The identification of mutation sites using breseq2 highlighted that both mutations and large chromosomal deletions can lead to the loss of LPS on the bacterial surface and that phage resistance mechanisms are quite easy to evolve since the mutations in any of the genes involved in LPS biosynthesis might lead to phage resistance, and the frequency is considerably high.

In conclusion, this case study substantiates the safety of phage therapy for BTIs and its capacity to alleviate infection symptoms, thus pioneering new avenues in therapeutic approaches. For chronic and complex infections, bacteriophages offer an economical and safe alternative. This case also illustrates the various challenges encountered during bacteriophage therapy, emphasizing the need for further research to better understand and enhance the clinical utility of bacteriophage therapy.

MATERIALS AND METHODS

P. aeruginosa Clinical Isolates and Antibiotic Susceptibility Assay (AST)

Clinical strains were isolated from the patient's bile using blood agar plates and subsequently identified using matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS, Vitek MS, bioMérieux, Rhône, France), with the comparison exploited by IVD 3.0 database [26]. The antimicrobial susceptibility of *P. aeruginosa* was determined by an automated antimicrobial susceptibility testing instrument Vitek 2 (bioMérieux, Rhône, France) according to the manufacturer's instructions [27]. For each isolate, susceptibility cards specific for gram-negative bacteria (Vitek AST-N335; bioMérieux, Rhône, France) were utilized, and all the experiments were conducted in triplicate following the manufacturer's recommendations.

Preparation of Therapeutic Phage Solutions

The phage agent was prepared as previously described [28]. Phages against the patient's *P. aeruginosa* isolates were screened from the Bacteriophage Library (Shanghai Institute of Phage, Shanghai, China) using the method of spot test and the double agar overlay method [29]. The susceptibilities of the target bacteria to phages were measured by the efficiency of plating (EOP) assay as previously described [30]. In brief, the phages were cultured in LB medium, subjected to chloroform sterilization and then centrifuged at 4 °C at 12,000 rpm for 10 min. The resulting supernatant was subsequently filtered through a 0.22-micron membrane for 18 h. Following this, the phage solution underwent serial 10-fold dilutions, after which 2 µL were applied onto an agar plate seeded with bacteria in the logarithmic growth phase.

For the initial phase of treatment, we selected the phage candidates φPA-A60, φPA-A69, φPA-A70, and φPA-A78 for inclusion in our therapeutic cocktail. These candidates were selected due to their robust antibacterial activity, characterized by the formation of large and clear plaques in our assays. For the subsequent treatment course, we utilized a dsRNA phage, phiYY, the amplification process of which has been detailed elsewhere [13]. All therapeutic phage solutions, with a concentration of 10⁸ PFU/mL, were aliquoted and packaged following GMP standards at the GMP Workshop of Zhongshan Hospital Fudan University, Shanghai, China.

Genome Sequencing and Analysis of *P. aeruginosa* Clinical Strains

Genomic DNA of the W220606 strain was isolated employing the cetyltrimethylammonium bromide (CTAB) technique and subsequently sequenced using both Pacific Biosciences (California, USA) and Illumina Miseq sequencing (California, USA) platforms. Post-sequencing, stringent removal of adapter sequences, and data sifting ensued, followed by the assembly of clean reads. This was accomplished utilizing AdapterRemoval [31] and SOAPec software packages. The resultant fragments were pieced together by SPAdes [32], meticulously crafting scaffolds and contigs with the aid of A5-miseq. The long-read data procured from the Pacbio platform was independently assembled by Canu [33], then, all assembled results were combined to generate a whole genome sequence using pilon software [34].

The genomes of the other 7 isolates were subjected to deep sequencing via the Illumina Hiseq 2500 platform, amassing approximately 1 gigabase per sample. Mutations were identified in clean reads using Breseq [35] with W220606 as the reference genome. Then the genome similarity between the eight strains was estimated by Sourmash and FastANI [36,37]. First, we computed the Sourmash signatures for every bacterial genome with two basic parameters (scaled = 1000, *k*-mer = 31). Then, we calculated ANI values on the whole genome Fast All Sequences in A (FASTA) sequences for the eight strains, with all being compared pairwise. A matrix file was generated as output by FastANI and was subsequently imported into Excel and plotted.

Bacterial genomes are available in the National Center for Biotechnology Information (NCBI) with the accession numbers PRJNA1168078, PRJNA1168582, PRJNA1168583, PRJNA1168584, PRJNA1168081, PRJNA1168084, PRJNA1168092, and PRJNA1167819 for W221201, W230623, W230629, W230629R, W221205, W221205R, W230622, and W220606, respectively. The sequences of the four dsDNA phages (φPA-A60, φPA-A69, φPA-A70, and φPA-A78) were deposited in the NCBI database with the accession number of NC_019913.1, PQ859842, PQ859843, and PQ859844, respectively. The genomes of phiYY were reported previously [17].

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DECLARATION OF COMPETING INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

ETHICS APPROVAL

This study was approved by the Ethics Committee of Shanghai Public Health Clinical Center (2020-S15701) and registered in the Chinese Clinical Trial Registry (ChiCTR1900020989). The patient and her legal guardian provided informed consent for phage

therapy and for using samples and data in research and publication.

AUTHOR CONTRIBUTIONS

Na Li: writing – original draft. **Linlin Li:** writing – original draft. **Bo He:** writing – original draft. **Dandan Li:** conceptualization. **Wenting Jin:** data curation. **Yuan Wu:** data curation. **Beidi Zhu:** data curation. **Mengjun Cheng:** formal analysis. **Nannan Wu:** conceptualization. **Demeng Tan:** conceptualization. **Jue Pan:** investigation. **Chunmei Zhou:** investigation. **Rong Bao:** investigation. **Hao Wu:** investigation. **Wen Zhang:** conceptualization. **Ming Li:** conceptualization. **Zhuojun Zhong:** formal analysis. **Jiazhen Liu:** writing – review & editing. **Jianglin Liao:** formal analysis. **Tongyu Zhu:** writing – review & editing. **Bijie Hu:** writing – review & editing. **Shuai Le:** conceptualization.

SUPPLEMENTARY DATA

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.hlife.2025.03.004>.

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